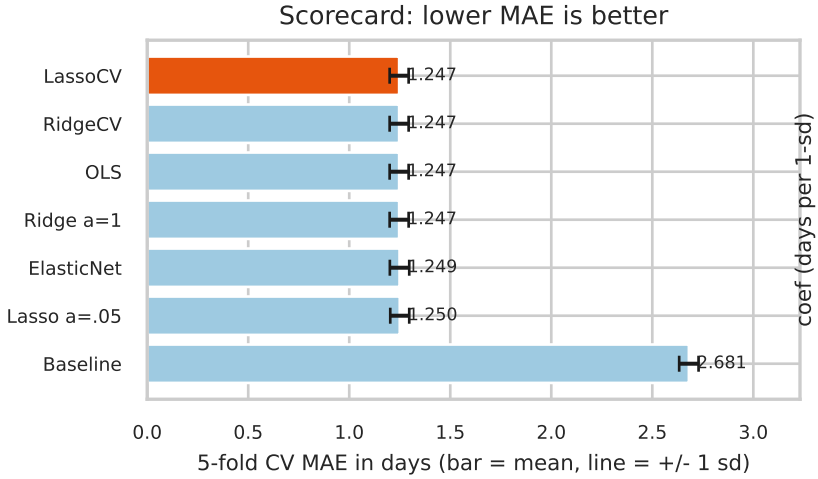
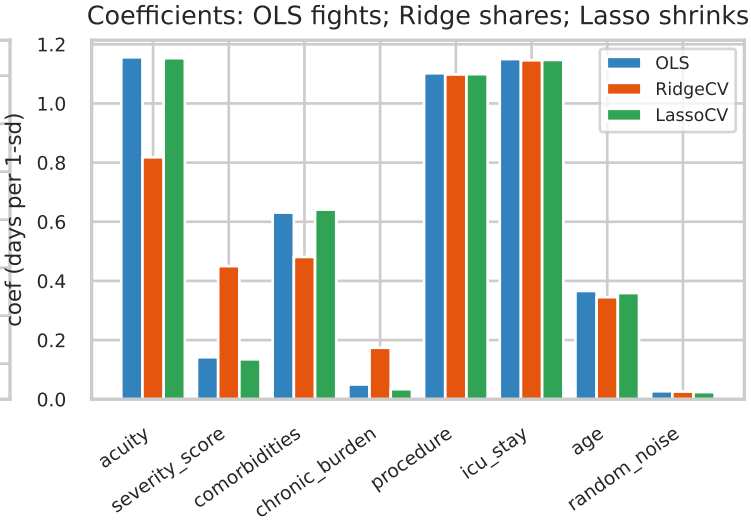
# Day 23 Cheat Sheet: Ridge and Lasso on LOS

Week 4 Day 2: shrink, zero, and read stable coefficients under multicollinearity

| TERM | WHAT IT MEANS | CODE |
|---|---|---|
| Multicollinearity | near-duplicate features fight for the same signal | acuity ~ severity_score |
| StandardScaler | always scale before L1/L2; penalties assume unit scale | StandardScaler() |
| Ridge (L2) | shrinks ALL coefs toward 0; keeps every feature | Ridge(alpha=...) |
| Lasso (L1) | can ZERO redundant echoes; sparse feature list | Lasso(alpha=...) |
| ElasticNet | blend of L1 and L2 (l1_ratio) | ElasticNet(alpha, l1_ratio) |
| RidgeCV / LassoCV | pick alpha on TRAIN by MAE (inner CV) | RidgeCV(alphas=..., scoring=) |
| Alpha | larger alpha = more shrinkage / more zeros | np.logspace(-3, 2, 40) |
| Stability rule | lowest CV MAE; prefer regularized within 1 fold sd | write BEFORE opening test |
| Hold-out test | opened once, for the champion only | final.predict(X_test) |




THIS DATA: 4,000 simulated discharges (seed 23), 12 features with clinical echoes, continuous los_days, 75/25 split
Mean LOS on train: 6.37 days (sd 3.28). Peak |corr| among echoes: 0.998
acuity~severity_score corr 0.998 | comorbidities~chronic_burden 0.995
RidgeCV alpha=12.66 | LassoCV alpha=0.003257
5-fold CV MAE: OLS 1.247, RidgeCV 1.247, LassoCV 1.247, baseline 2.681
Champion: Lasso (CV alpha). Test once: MAE 1.301 days, RMSE 1.633, R2 0.766
Within 2 days: 79% of test patients. Median |error| 1.08 days.

HABITS THAT KEEP YOU HONEST
- Always StandardScaler before Ridge/Lasso/ElasticNet; penalties assume comparable scales
- Tune alpha on TRAIN with RidgeCV/LassoCV by MAE; never peek at the test set to pick alpha
- Read coefficient paths: OLS fights on near-duplicates; Ridge shares credit; Lasso can zero echoes
- Prefer regularized within 1 fold sd of the best CV MAE (stability over raw OLS bravado)
- Write the selection rule before opening test; report MAE/RMSE in days alongside R2

```
ridge_cv = make_pipeline(StandardScaler(),
    RidgeCV(alphas=np.logspace(-3,2,40), scoring="neg_mean_absolute_error", cv=cv))
lasso_cv = make_pipeline(StandardScaler(),
    LassoCV(alphas=np.logspace(-3,2,40), cv=cv, max_iter=20000, random_state=23))
```

# Infographic: When clinical features echo each other

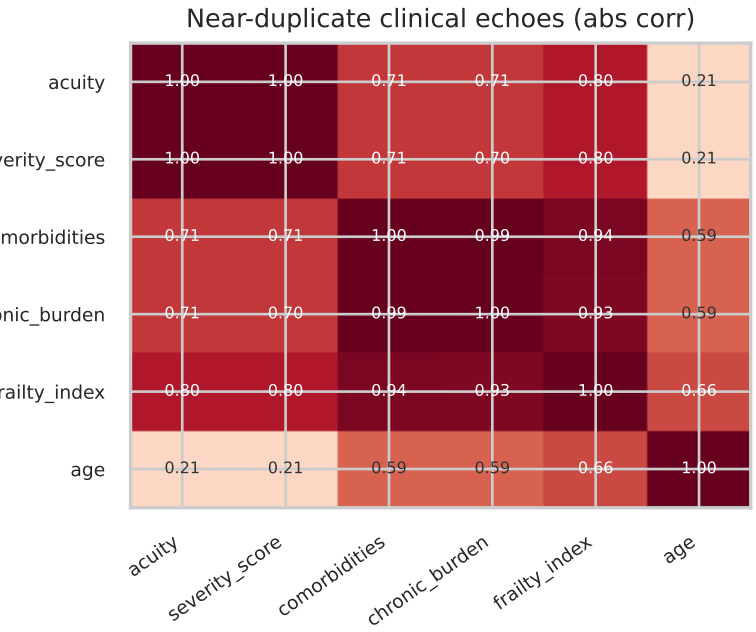
OLS fights; Ridge shrinks all; Lasso can zero; pick alpha by CV MAE on train

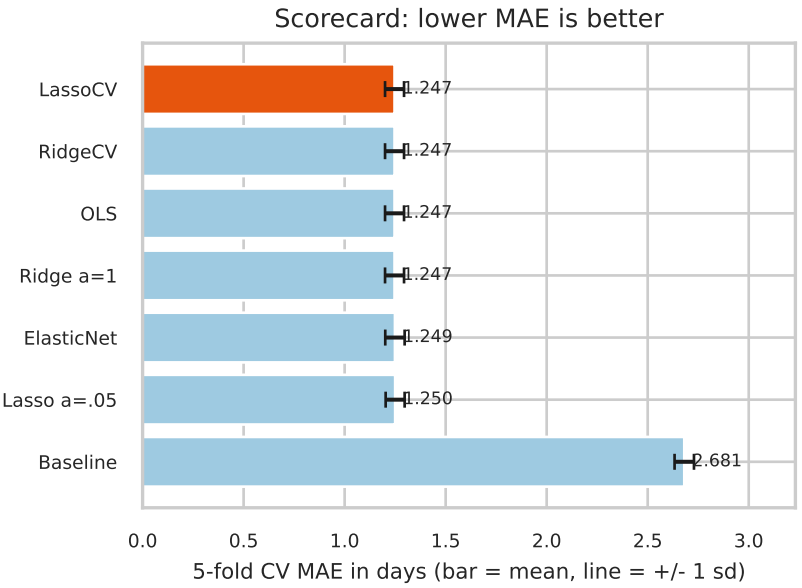
**CHAMPION**
**LassoCV**

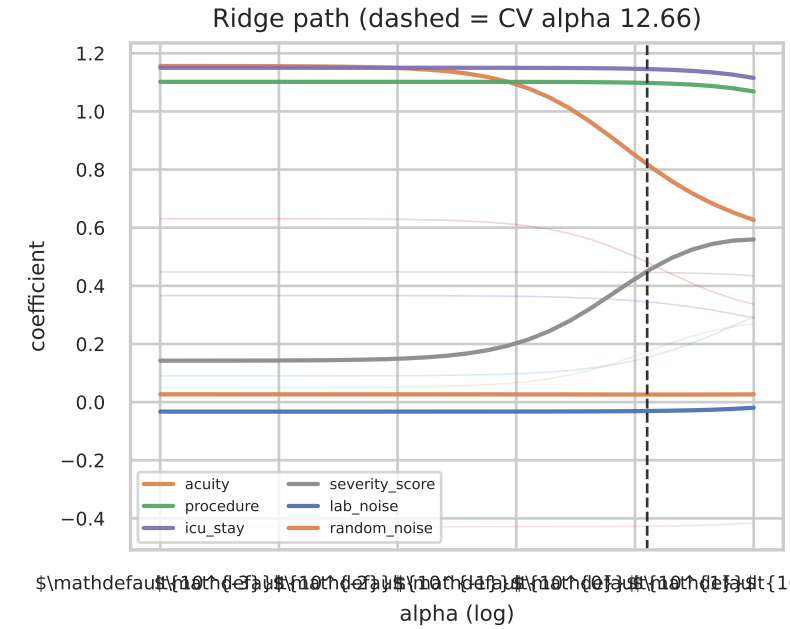
**TEST MAE**
**1.30 d**

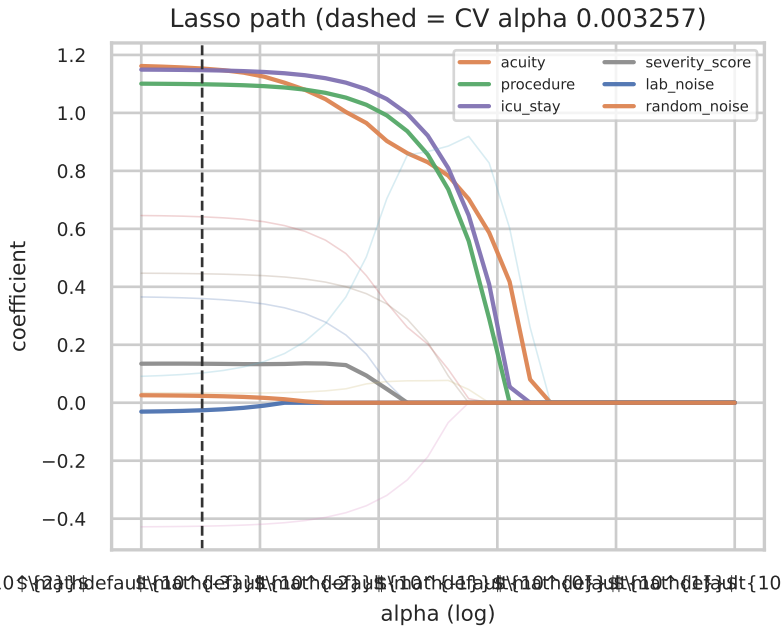
**RIDGE a**
**12.66**

**LASSO a**
**0.003257**




Takeaways
- Near-duplicates (acuity ~ severity_score corr ~0.998) make OLS coefficients fight for the same signal.
- Ridge (L2) shrinks every coefficient; acuity and severity share credit more evenly than OLS.
- Lasso (L1) can zero redundant echoes at larger alpha; CV here chose a mild alpha, so zeros come later on the path.
- Always scale first. Pick alpha with RidgeCV/LassoCV on TRAIN by MAE. Prefer regularized within 1 fold sd.
- Tomorrow: feature interactions and polynomial terms for length of stay.

# Hands-On Worksheet: ds-genai-90-days / day23

60-90 minutes | notebook: day23/ridge_lasso_los.ipynb + day23/README.md

[ ] mkdir day23, conda activate ds90, open day23/ridge_lasso_los.ipynb

[ ] Simulate ~4,000 discharges with clinical echoes (snippet 1, seed 23)

[ ] Confirm peak |corr| among acuity/severity and comorbidities/chronic_burden

[ ] Define models: Dummy, OLS+scaler, Ridge, Lasso, ElasticNet (snippet 2)

[ ] Run 5-fold CV MAE/RMSE/R2; then RidgeCV/LassoCV for alpha (snippet 3)

[ ] Record RidgeCV alpha _____ and LassoCV alpha _____

[ ] Compare OLS vs RidgeCV vs LassoCV coefficients (snippet 4)

[ ] Note which features Lasso zeros (if any) at the CV alpha

[ ] Write selection rule BEFORE test: lowest CV MAE, prefer regularized within 1 sd

[ ] Pick champion, score test once, print MAE/RMSE/R2 and residuals

[ ] Plot Ridge and Lasso coefficient paths vs alpha (snippet 5)

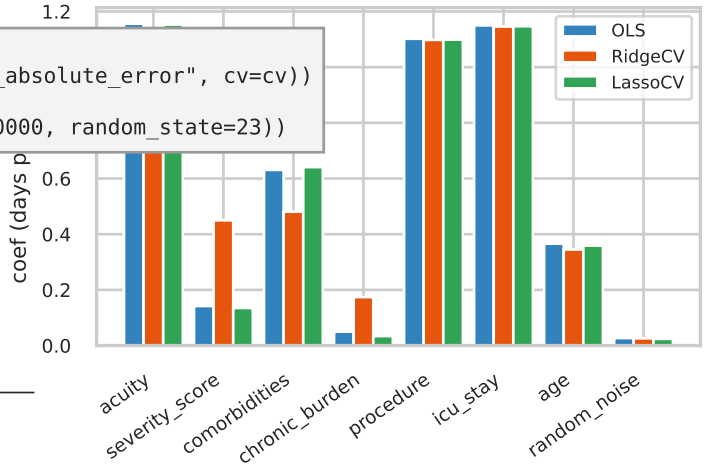
[ ] README: scorecard, coef comparison, champion and why, limits

[ ] git add day23 && git commit && git push (snippet 6)

[ ] Stretch: Snowflake AVG/ABS error on scored predictions (snippet 7)

```
ridge_cv = make_pipeline(StandardScaler(),
    RidgeCV(alphas=np.logspace(-3,2,40), scoring="neg_mean_absolute_error", cv=cv))
lasso_cv = make_pipeline(StandardScaler(),
    LassoCV(alphas=np.logspace(-3,2,40), cv=cv, max_iter=20000, random_state=23))
```

Coefficients: OLS fights; Ridge shares; Lasso shrinks




CV MAE OLS: _____ RidgeCV: _____ LassoCV: _____ baseline: _____

Champion: _____ Test MAE: _____ d RMSE: _____ R2: _____

# Day 23 Self-Check Quiz

1. Why do OLS coefficients fight when acuity and severity_score are near-duplicates?
   _____

2. What does Ridge (L2) do to coefficients that OLS does not?
   _____

3. What can Lasso (L1) do that Ridge cannot, and why does that help care managers?
   _____

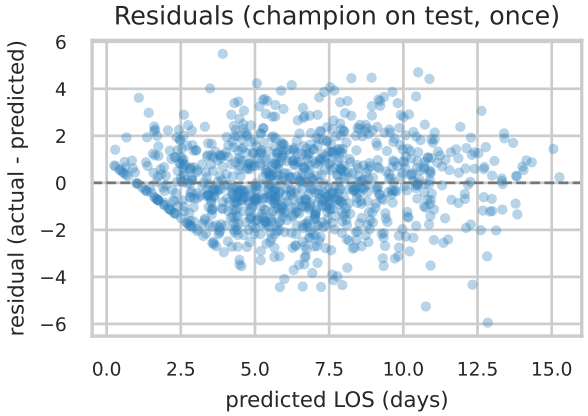
4. Why must you StandardScaler before Ridge or Lasso?
   _____

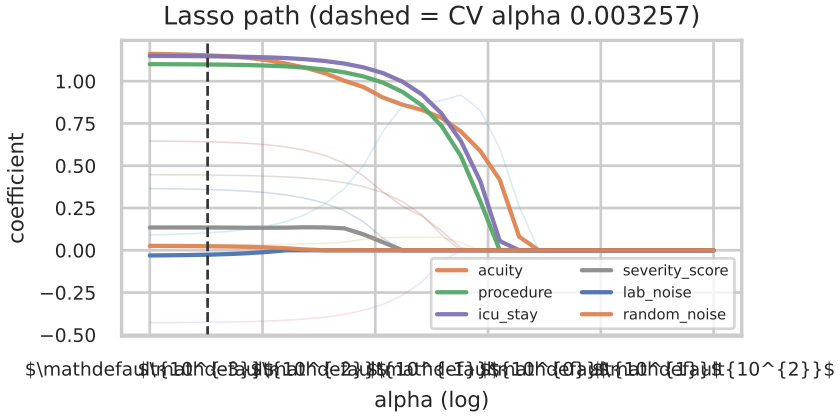
5. Where should you pick alpha: train CV or the hold-out test set?
   _____

6. What is the stability selection rule used today?
   _____

7. Looking at a Lasso path, what happens to coefficients as alpha grows large?
   _____

8. Why report MAE in days alongside R2 even after regularization?
   _____




Answers:
1 they measure almost the same signal, so OLS can assign an arbitrary split (one large, one small or flipped)
2 Ridge shrinks ALL coefficients toward zero, sharing credit more evenly across correlated features
3 Lasso can zero redundant echoes, giving a shorter feature list that is easier to explain at the bedside
4 L1/L2 penalties assume features are on a comparable scale; unscaled age would dominate acuity
5 on TRAIN with RidgeCV/LassoCV (inner CV by MAE); never use the hold-out to tune alpha
6 lowest CV MAE, then prefer LassoCV/RidgeCV/ElasticNet if within 1 fold sd of the best
7 more coefficients hit exactly zero; at the largest alpha here, Lasso zeros all 12 features
8 clinicians hear days off, not fractions of variance; regularization does not change that habit